Lab 3 - Routing Protocol

NAME: POH LOK YEE

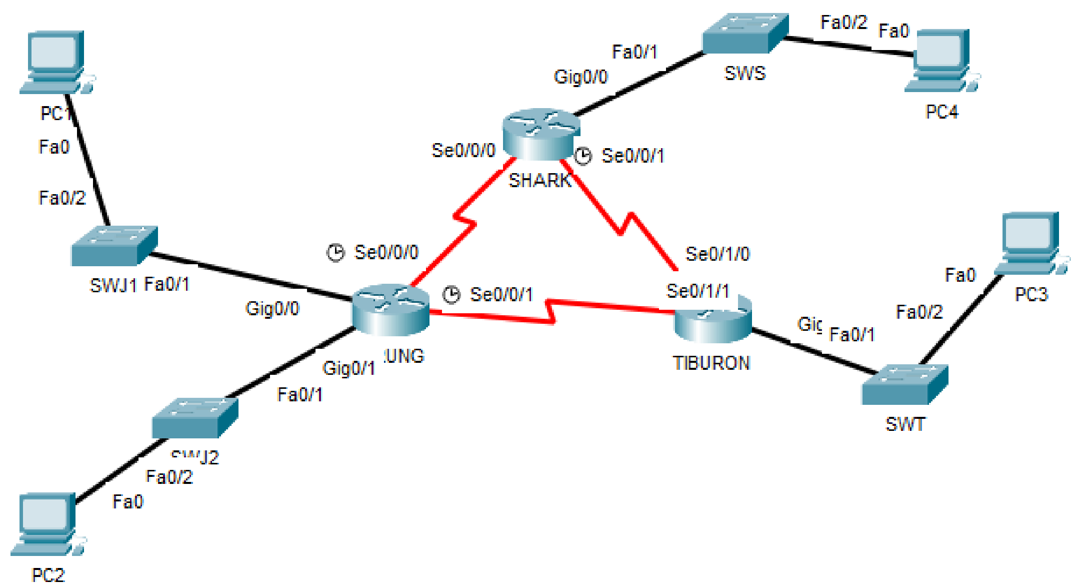
MATRIC NUM.: A23CS0262

SECTION: 12

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
K8Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

=

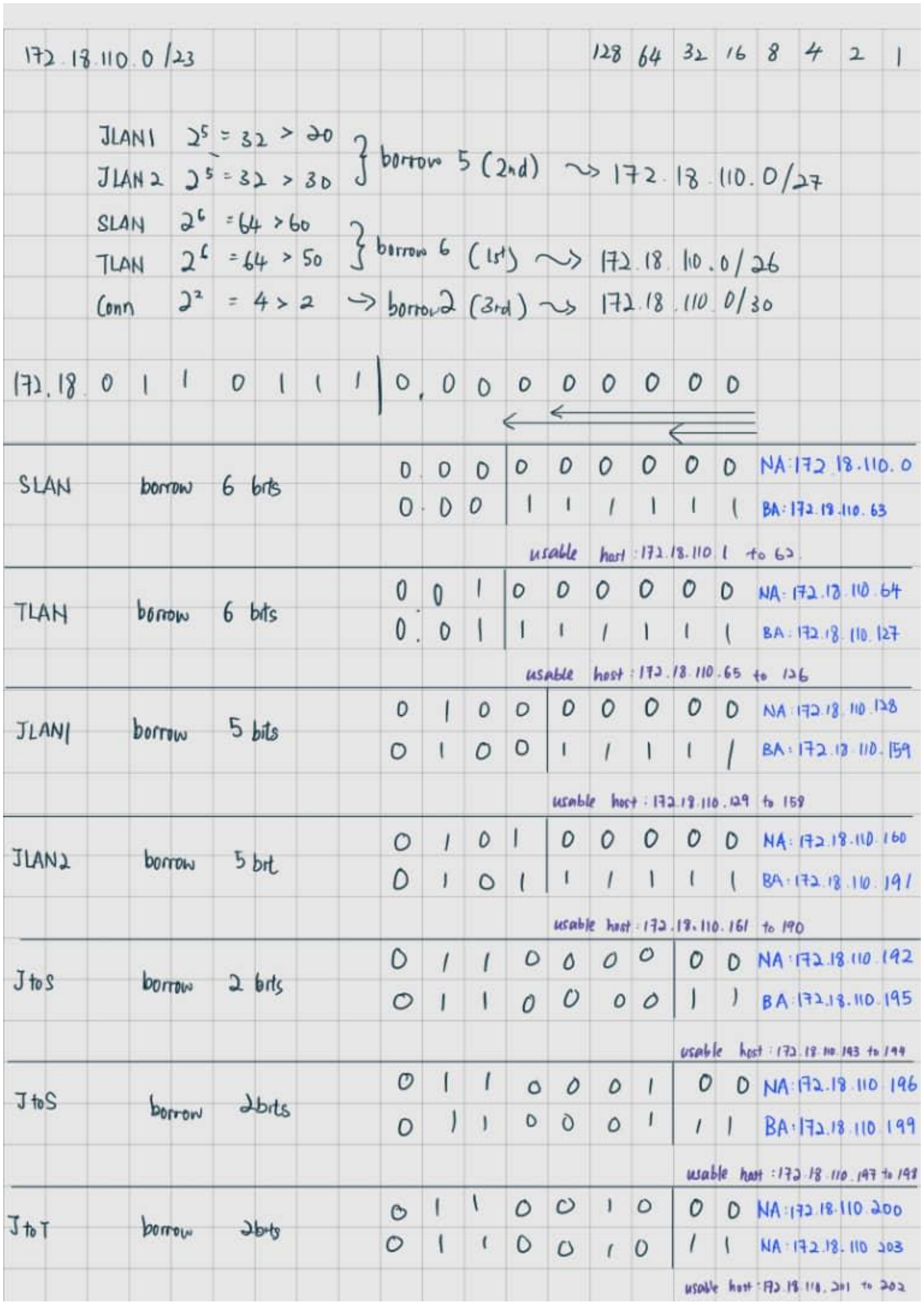
Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	-
2		Se0/0/1	172.18.110.197	255.255.255.252	-
3		G0/0	172.18.110.129	255.255.255.224	-
4		G0/1	172.18.110.161	255.255.255.224	-
5	TIBURON	Se0/1/1-b	172.18.110.198	255.255.255.252	-
6		Se0/1/0	172.18.110.202	255.255.255.252	-
7		G0/0	172.18.110.65	255.255.255.192	-
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	-
9		Se0/0/1	172.18.110.201	255.255.255.252	-
10		G0/0	172.18.110.1	255.255.255.192	-
11	PC1	-	172.18.110.158	255.255.255.224	172.18.110.129
12	PC2	-	172.18.110.190	255.255.255.224	172.18.110.161
13	PC3	-	172.18.110.126	255.255.255.192	172.18.110.65
14	PC4	-	172.18.110.62	255.255.255.192	172.18.110.1

LAB TASKS

Task 1 – IP Addressing

1. Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.



172.18.110.0 /23 — 128 64 32 16 8 4 2 1

JLAN1 $2^5 = 32 > 20$
JLAN2 $2^5 = 32 > 30$ } borrow 5 (2nd) ~> 172.18.110.0/27

SLAN $2^6 = 64 > 60$
TLAN $2^6 = 64 > 50$ } borrow 6 (1st) ~> 172.18.110.0/26

Conn $2^2 = 4 > 2$ → borrow 2 (3rd) ~> 172.18.110.0/30

172.18 0 1 1 0 1 1 1 | 0. 0 0 0 0 0 0 0 0

SLAN	borrow	6 bits	0	0	0	0	0	0	0	0	0	NA:172.18.110.0
			0	0	0	1	1	1	1	1	1	BA:172.18.110.63
						usable host: 172.18.110.1 to 62						
TLAN	borrow	6 bits	0	0	1	0	0	0	0	0	0	NA: 172.18.110.64
			0	0	1	1	1	1	1	1	1	BA: 172.18.110.127
						usable host: 172.18.110.65 to 126						
JLAN1	borrow	5 bits	0	1	0	0	0	0	0	0	0	NA: 172.18.110.128
			0	1	0	0	1	1	1	1	1	BA: 172.18.110.159
						usable host: 172.18.110.129 to 158						
JLAN2	borrow	5 bit	0	1	0	1	0	0	0	0	0	NA: 172.18.110.160
			0	1	0	1	1	1	1	1	1	BA: 172.18.110.191
						usable host: 172.18.110.161 to 190						
J to S	borrow	2 bits	0	1	1	0	0	0	0	0	0	NA: 172.18.110.192
			0	1	1	0	0	0	0	1	1	BA: 172.18.110.195
									usable host: 172.18.110.193 to 194			
J to S	borrow	2bits	0	1	1	0	0	0	1	0	0	NA: 172.18.110.196
			0	1	1	0	0	0	1	1	1	BA: 172.18.110.199
									usable host: 172.18.110.197 to 198			
J to T	borrow	2bits	0	1	1	0	0	1	0	0	0	NA: 172.18.110.200
			0	1	1	0	0	1	0	1	1	NA: 172.18.110.203
									usable host: 172.18.110.201 to 202			

2. Using the IP addresses calculated, configure the devices with the appropriate information.

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.
    a. CLI
        i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

JERUNG:

```
JERUNG#
%SYS-5-CONFIG_I: Configured from console by console
enable
JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.18.110.128/27 is directly connected, GigabitEthernet0/0
L       172.18.110.129/32 is directly connected, GigabitEthernet0/0
C       172.18.110.160/27 is directly connected, GigabitEthernet0/1
L       172.18.110.161/32 is directly connected, GigabitEthernet0/1
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.193/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1

JERUNG#
```

TIBURON:

```
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0

TIBURON#
```

SHARK:

```
SHARK#
%SYS-5-CONFIG_I: Configured from console by console
enable
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1

SHARK#
```

Jerung

b. PT tool

i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

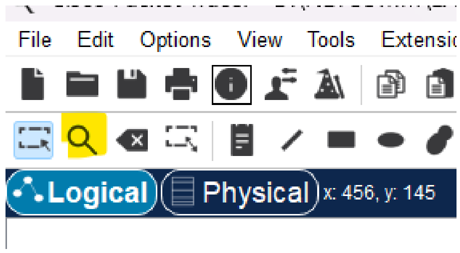


Figure B

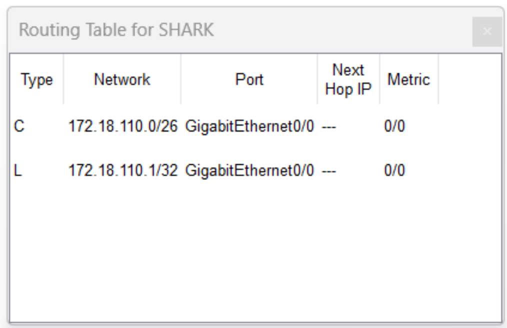


Figure C

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.160/27	GigabitEthernet0/1	---	0/0
L	172.18.110.161/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

2. Try to ping PC2 from PC1, paste the results here.

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.190: bytes=32 time<1ms TTL=127
Reply from 172.18.110.190: bytes=32 time<1ms TTL=127
Reply from 172.18.110.190: bytes=32 time<1ms TTL=127

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Success

3. Try to ping PC4 from PC1, paste the results here.

```
Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Failed

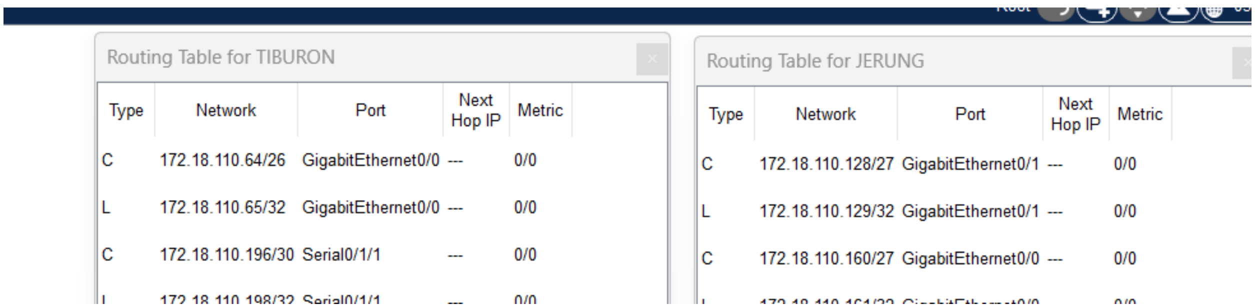
4. Explain the reason(s) behind the results.

PC2 ping from PC1 successfully as they are in a same subnet and router. PC4 cannot ping from PC1 as they are not in a same subnet and router. Different router cannot communicate with each other. The network between them are not configure.

5. What needs to be done to ensure all PCs can ping each other successfully?
   The router must be properly configured to interconnect all network segments for ensuring a seamless routing path for traffic between the PCs.

Task 2 – Routing Configuration

1. Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.



Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.160/27	GigabitEthernet0/0	---	0/0

Figure D

2. In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```
JERUNG#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#
```

Figure E

   a. What can you say about the addresses used in the 'network' instructions in Figure E?
      The addresses used are all network address (NA) of router JERUNG's. They act as the router information (self-contained network) of JERUNG.

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```
TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200
```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here

JERUNG:

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.64/26	Serial0/0/1	172.18.110.198	120/1
C	172.18.110.128/27	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.160/27	GigabitEthernet0/1	---	0/0
L	172.18.110.161/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1

```
JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 10 subnets, 4 masks
R       172.18.110.64/26 [120/1] via 172.18.110.198, 00:00:01, Serial0/0/1
C       172.18.110.128/27 is directly connected, GigabitEthernet0/0
L       172.18.110.129/32 is directly connected, GigabitEthernet0/0
C       172.18.110.160/27 is directly connected, GigabitEthernet0/1
L       172.18.110.161/32 is directly connected, GigabitEthernet0/1
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.193/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1
R       172.18.110.200/30 [120/1] via 172.18.110.198, 00:00:01, Serial0/0/1

JERUNG#
```

TIBURON:

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.160/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

```
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 9 subnets, 4 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
R       172.18.110.128/27 [120/1] via 172.18.110.197, 00:00:08, Serial0/1/1
R       172.18.110.160/27 [120/1] via 172.18.110.197, 00:00:08, Serial0/1/1
R       172.18.110.192/30 [120/1] via 172.18.110.197, 00:00:08, Serial0/1/1
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0

TIBURON#
```

a. What are the changes seen in TIBURON?
   A new route generated by the RIP protocol shown in the routing table and marked with an “R” which stands for RIP.

b. What are the Networks with type R in TIBURON and JERUNG?
   JERUNG:
   - 172.18.110.64/26
   - 172.18.110.200/30

   TIBURON:
   - 172.18.110.128/27
   - 172.18.110.160/27
   - 172.18.110. 192/30

c. Ping PC3 from PC1. Was it successful?

```
Pinging 172.18.110.126 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.126: bytes=32 time=10ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126
Reply from 172.18.110.126: bytes=32 time=9ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 10ms, Average = 6ms
```

Success

d. Ping PC4 from PC1. Was it successful?

```
Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

No. It failed

e. Explain the reasons for your answer.

PC1 pings PC3 successfully as both are in the same subnet and they are directly connected to JERUNG, the network between them is correctly configured. The RIP routing protocols on JERUNG(PC 1) and TIBURON(PC 3) have been automatically updated in the routers' routing tables.

PC1 fail to pings PC4 due to an improperly configured network connection and the absence of a set route. Additionally, the RIP routing protocol and routing table on SHARK (for PC4) have not been configured correctly.

f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK#
SHARK#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.0
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.200
SHARK(config-router)#no auto-summary
SHARK(config-router)#
```

g. Open router SHARK's routing table and paste here.

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
R	172.18.110.64/26	Serial0/0/1	172.18.110.202	120/1
R	172.18.110.128/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.160/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 10 subnets, 4 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
R       172.18.110.64/26 [120/1] via 172.18.110.202, 00:00:02, Serial0/0/1
R       172.18.110.128/27 [120/1] via 172.18.110.193, 00:00:03, Serial0/0/0
R       172.18.110.160/27 [120/1] via 172.18.110.193, 00:00:03, Serial0/0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
R       172.18.110.196/30 [120/1] via 172.18.110.193, 00:00:03, Serial0/0/0
                          [120/1] via 172.18.110.202, 00:00:02, Serial0/0/1
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1
 --More--
```

h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result
1	Cisco Packet Tracer PC Command Line 1.0 C:\>ping 172.18.110.158 Pinging 172.18.110.158 with 32 bytes of data: Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Reply from 172.18.110.158: bytes=32 time=11ms TTL=126 Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.158: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 11ms, Average = 3ms
2	C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 1ms, Average = 1ms
3	C:\>ping 172.18.110.126 Pinging 172.18.110.126 with 32 bytes of data: Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Reply from 172.18.110.126: bytes=32 time=8ms TTL=126 Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.126: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 8ms, Average = 2ms C:\>

Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
   a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: 192.168.1.0/24
   b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.254
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1

   c. After this change, can PC4 and PC1 ping PC3?

```
C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

PC4 ping PC3 is failed.

```
C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Request timed out.
Reply from 172.18.110.129: Destination host unreachable.

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

PC1 ping PC3 is failed.

d. Copy and paste the routing tables for both SHARK and TIBURON here.

TIBURON:

```
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 8 subnets, 4 masks
R       172.18.110.0/26 [120/1] via 172.18.110.201, 00:00:06, Serial0/1/0
R       172.18.110.128/27 [120/1] via 172.18.110.197, 00:00:04, Serial0/1/1
R       172.18.110.160/27 [120/1] via 172.18.110.197, 00:00:04, Serial0/1/1
R       172.18.110.192/30 [120/1] via 172.18.110.197, 00:00:04, Serial0/1/1
                          [120/1] via 172.18.110.201, 00:00:06, Serial0/1/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
 --More--
```

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/26	Serial0/1/0	172.18.110.201	120/1
R	172.18.110.128/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.160/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0
C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

SHARK:

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.18.0.0/16 is variably subnetted, 10 subnets, 4 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
R       172.18.110.64/26 is possibly down, routing via 172.18.110.202, Serial0/0/1
R       172.18.110.128/27 [120/1] via 172.18.110.193, 00:00:10, Serial0/0/0
R       172.18.110.160/27 [120/1] via 172.18.110.193, 00:00:10, Serial0/0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
R       172.18.110.196/30 [120/1] via 172.18.110.193, 00:00:10, Serial0/0/0
                          [120/1] via 172.18.110.202, 00:00:10, Serial0/0/1
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1
 --More--
```

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
R	172.18.110.64/26	Serial0/0/1	172.18.110.202	120/16
R	172.18.110.128/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.160/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

e. Referring to the routing table, explain your findings.
   The routing table has not been updated with the new network address of the subnet. Packets intended for PC3 are dropped, leading to failed ping attempts from PC4 and PC1 to PC3 after the IP address change.

f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?
   Update the routing configurations on TIBURON to include the new subnet (192.168.1.0/24) and enable its advertisement through the RIP protocol. This will ensure the subnet is automatically propagated and updated in TIBURON’s routing table, enabling connectivity between all PCs.

g. Show your configurations in TIBURON to ensure end-to-end connectivity.

```
TIBURON#
TIBURON#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#network 172.18.110.196
TIBURON(config-router)#network 172.18.110.200
TIBURON(config-router)#no auto-summary
TIBURON(config-router)#
```

h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

PC	Ping result
1	Cisco Packet Tracer PC Command Line 1.0 C:\>ping 172.18.110.158 Pinging 172.18.110.158 with 32 bytes of data: Reply from 172.18.110.158: bytes=32 time=2ms TTL=126 Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Reply from 172.18.110.158: bytes=32 time=2ms TTL=126 Ping statistics for 172.18.110.158: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 2ms, Average = 1ms
2	Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=21ms TTL=126 Reply from 172.18.110.190: bytes=32 time=12ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 21ms, Average = 8ms

4	C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Reply from 172.18.110.62: bytes=32 time=9ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=11ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 11ms, Average = 5ms

REFLECTION

What have you learned in this task?

In this lab, I developed a comprehensive understanding of subnetting and routing protocols. I gained practical experience in configuring IP addresses, subnet masks, and gateways to ensure device connectivity. Troubleshooting connectivity issues emphasized the need to verify routing configurations and propagate routes effectively. The use of Packet Tracer tools allowing me to visualize and optimize network topologies. This lab significantly enhanced my skills in network design and troubleshooting.